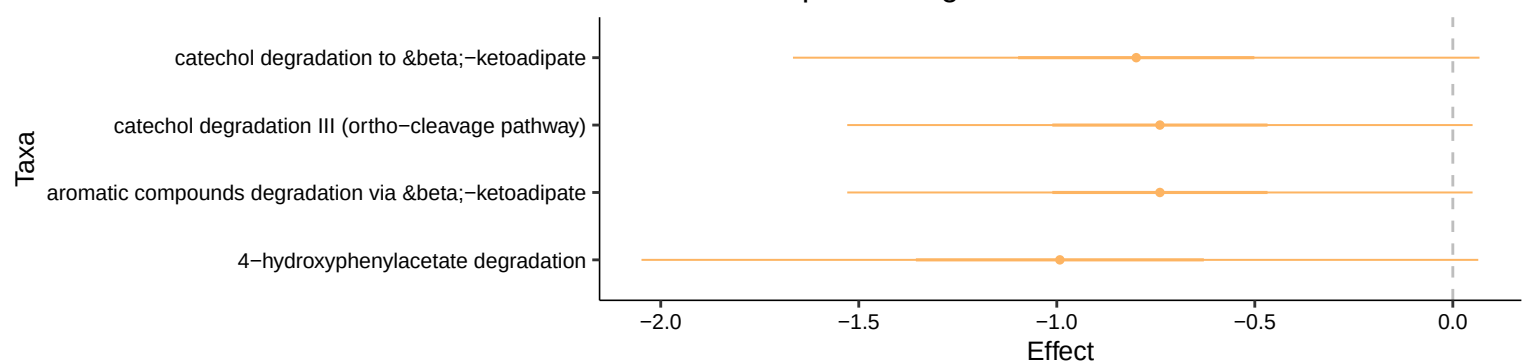
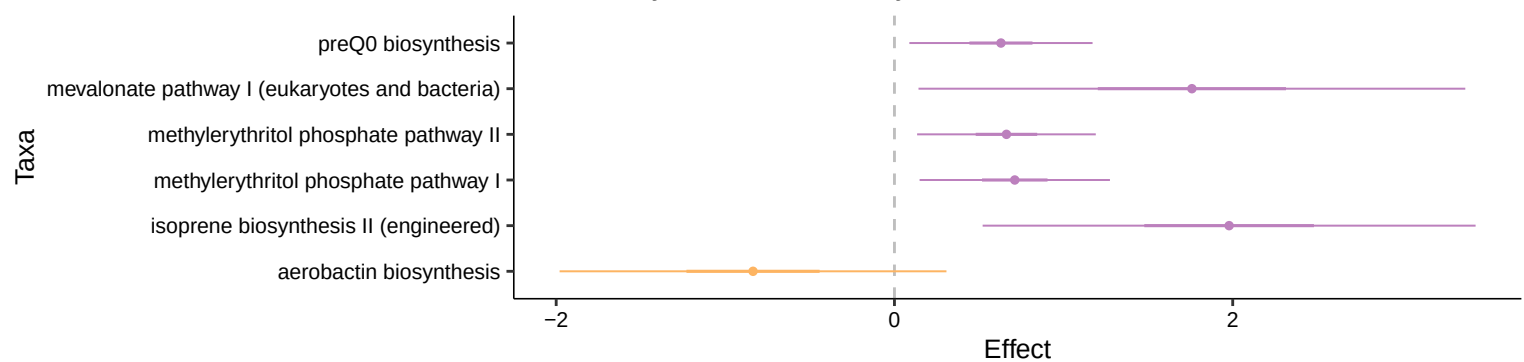


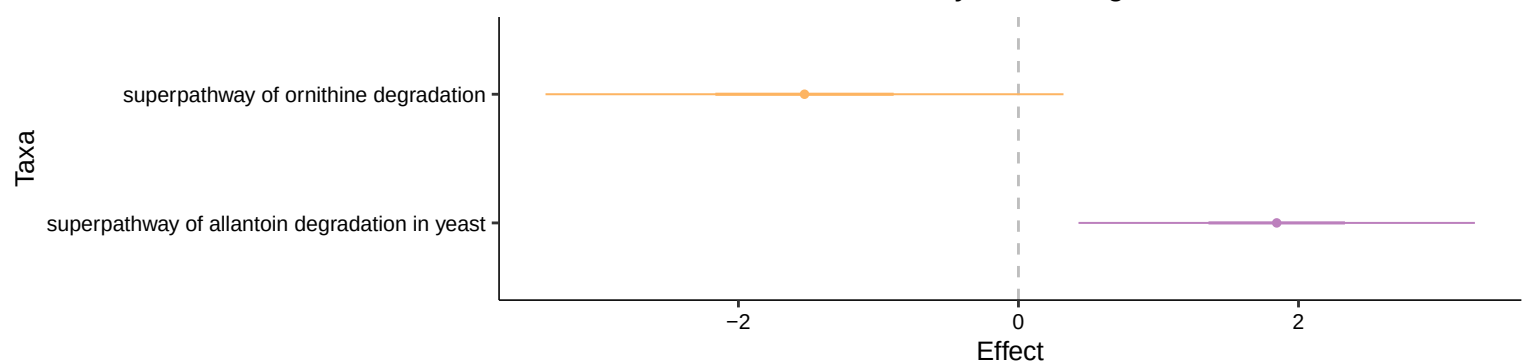
Aromatic Compound Degradation



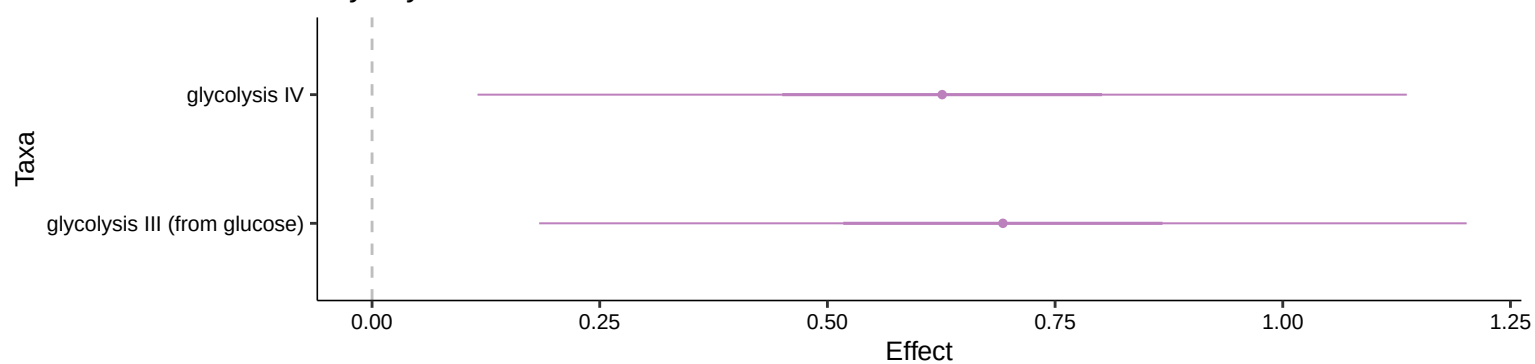
Secondary Metabolite Biosynthesis



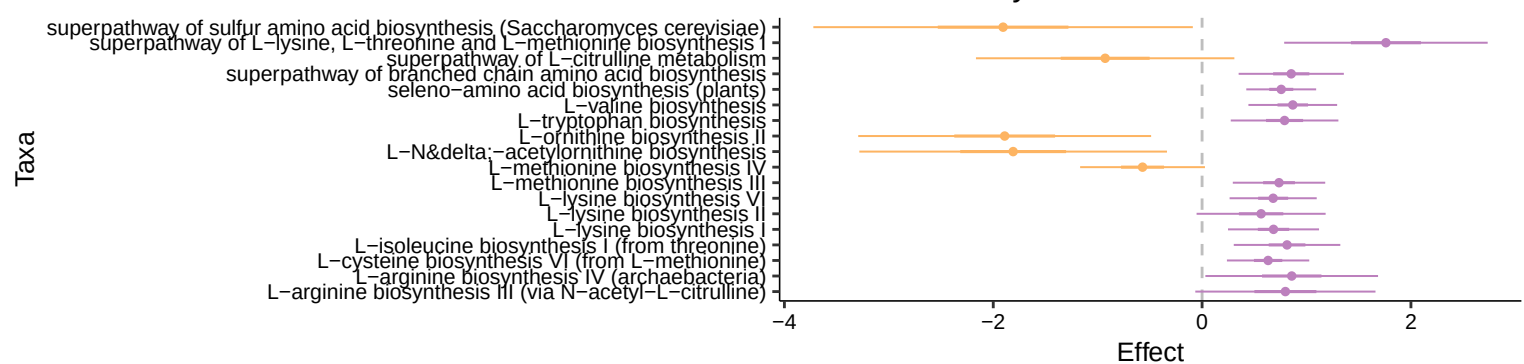
Amide, Amidine, Amine, and Polyamine Degradation



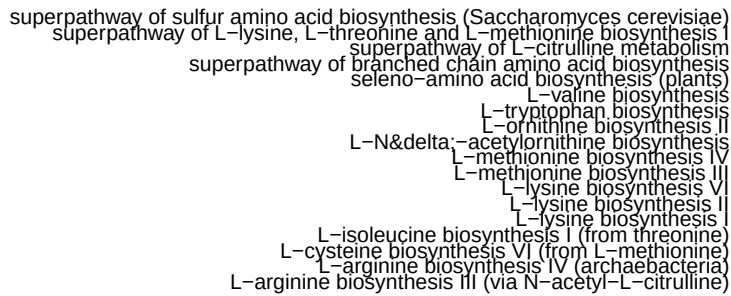
Glycolysis



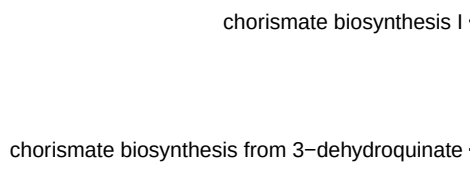
Amino Acid Biosynthesis



superpathway of sulfur amino acid biosynthesis (Saccharomyces cerevisiae)
 superpathway of L-lysine, L-threonine and L-methionine biosynthesis I
 superpathway of L-citrulline biosynthesis I
 superpathway of branched chain amino acid biosynthesis
 seleno-amino acid biosynthesis (plants)
 L-valine biosynthesis
 L-tryptophan biosynthesis
 L-ornithine biosynthesis II
 L-N δ -acetylornithine biosynthesis
 L-methionine biosynthesis IV
 L-methionine biosynthesis II
 L-lysine biosynthesis VI
 L-lysine biosynthesis I
 L-lysine biosynthesis I
 L-isoleucine biosynthesis I (from threonine)
 L-cysteine biosynthesis VI (from L-methionine)
 L-arginine biosynthesis IV (archaeobacteria)
 L-arginine biosynthesis III (via N-acetyl-L-citrulline)



chorismate biosynthesis from 3-dehydroquinate

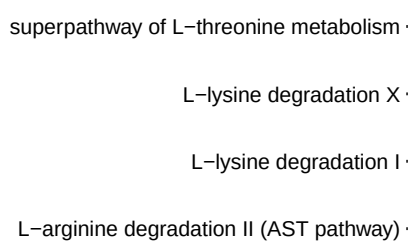


superpathway of L-threonine metabolism

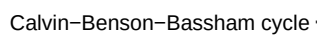
L-lysine degradation X

L-lysine degradation I

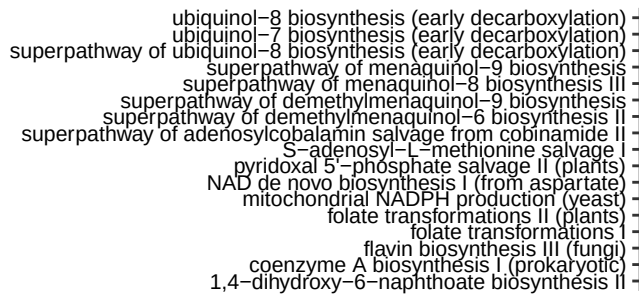
L-arginine degradation II (AST pathway)



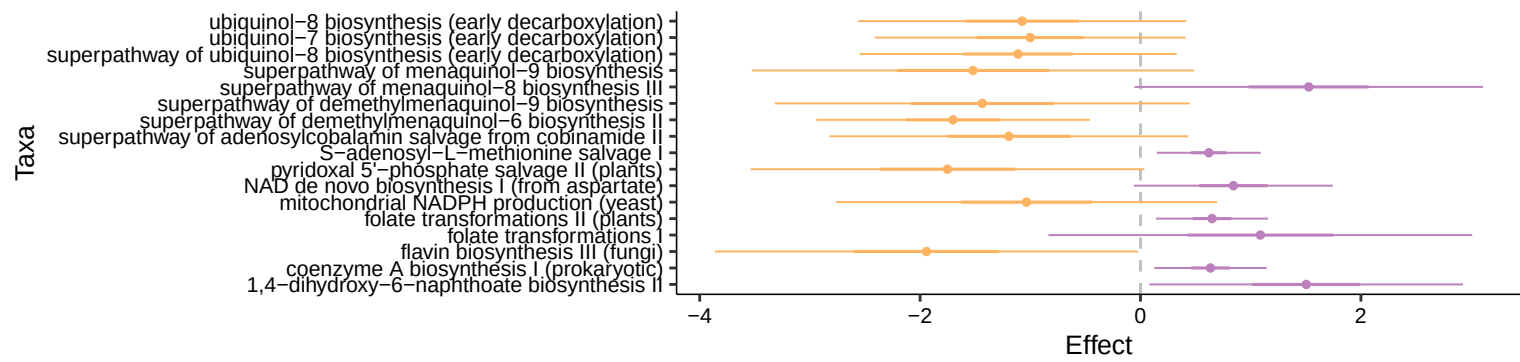
Calvin–Benson–Bassham cycle



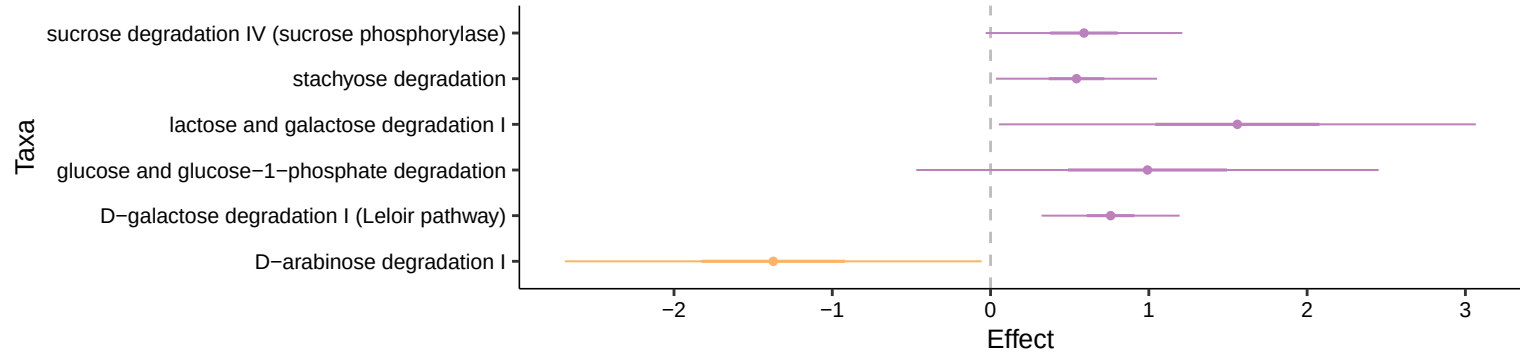
ubiquinol-8 biosynthesis (early decarboxylation)
 ubiquinol-8 biosynthesis (early decarboxylation)
 superpathway of ubiquinol-8 biosynthesis (early decarboxylation)
 superpathway of menaquinol-9 biosynthesis
 superpathway of menaquinol-8 biosynthesis III
 superpathway of demethylmenaquinol-9 biosynthesis
 superpathway of demethylmenaquinol-6 biosynthesis II
 superpathway of adenosylcobalamin salvage from cobinamide II
 S-adenosyl-L-methionine salvage
 pyridoxal 5'-phosphate salvage II (plants)
 NAD de novo biosynthesis I (from aspartate)
 mitochondrial NADPH production (yeast)
 folate transformations II (plants)
 folate transformations I
 flavin biosynthesis III (fungi)
 coenzyme A biosynthesis I (prokaryotic)
 1,4-dihydroxy-6-naphthoate biosynthesis II



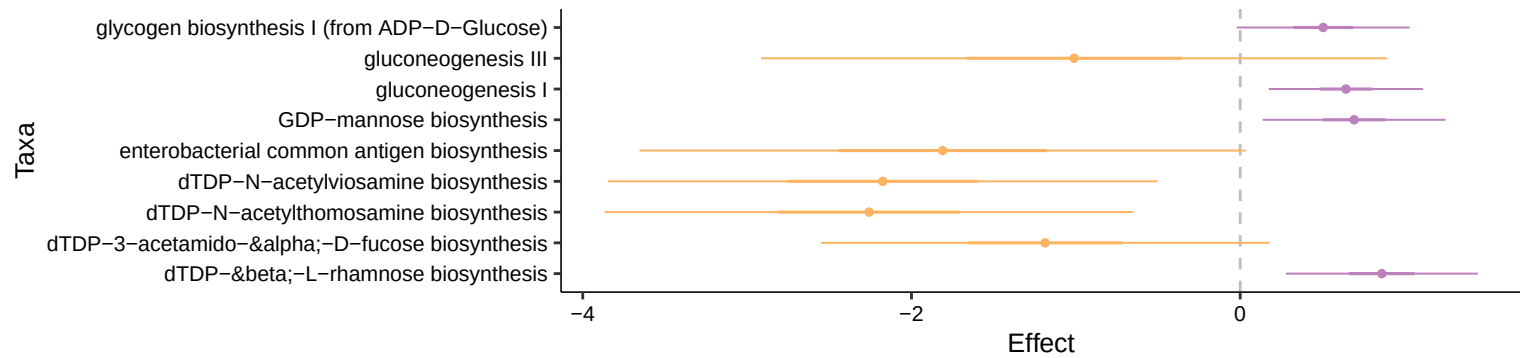
Cofactor, Carrier, and Vitamin Biosynthesis



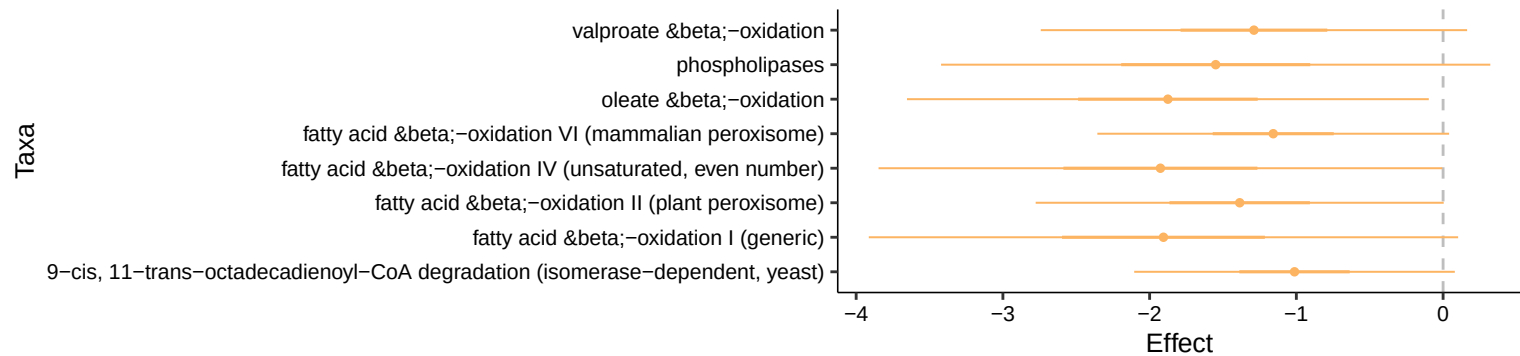
Carbohydrate Degradation



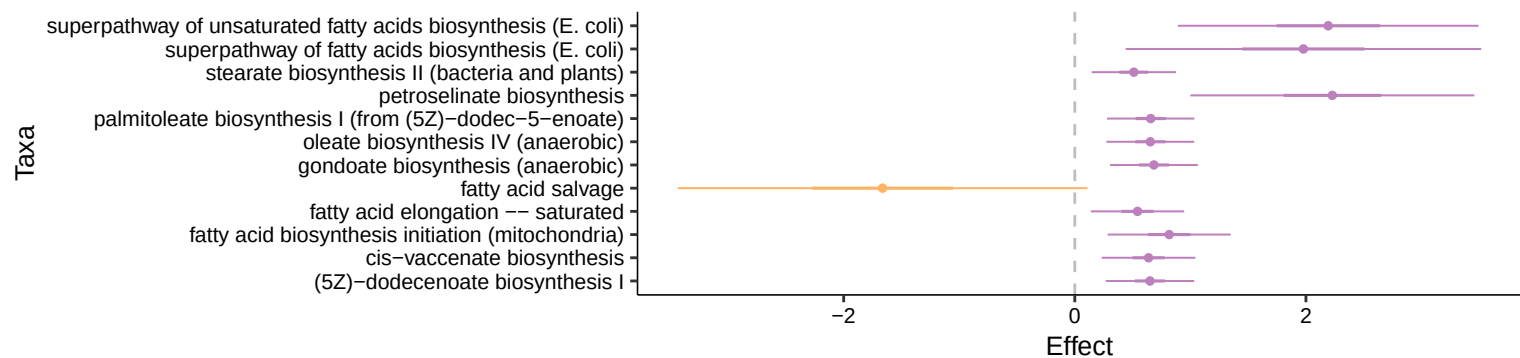
Carbohydrate Biosynthesis



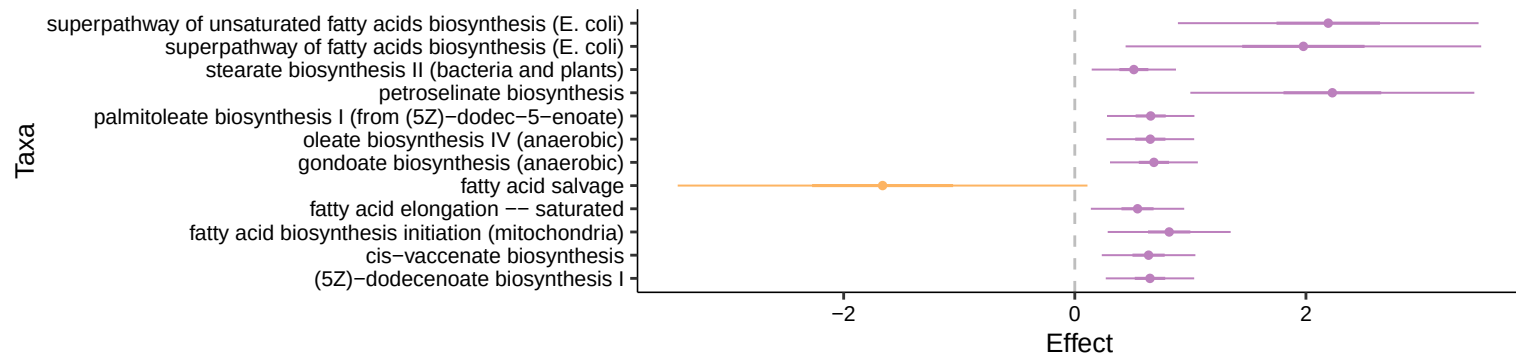
Fatty Acid and Lipid Degradation



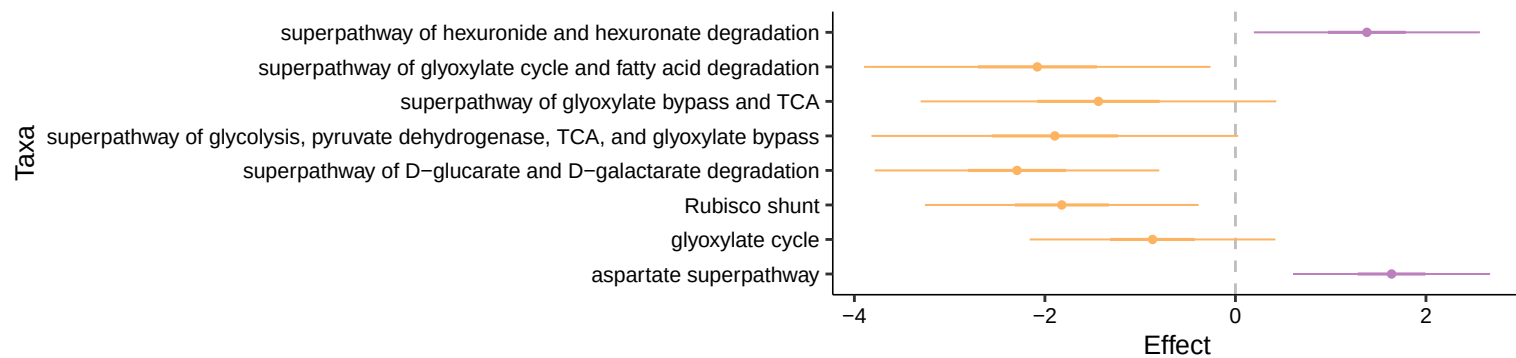
Fatty Acid and Lipid Biosynthesis



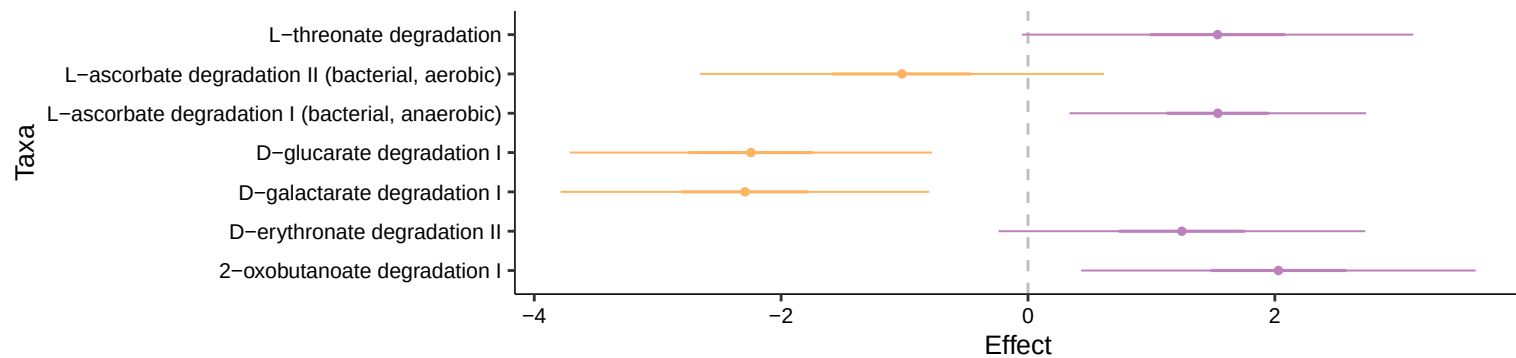
Fatty Acid and Lipid Biosynthesis



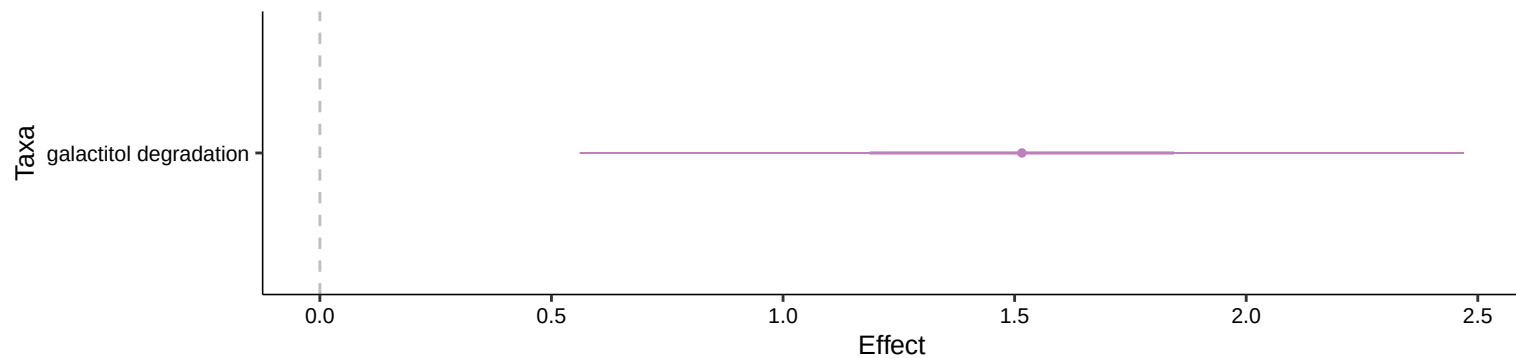
NA



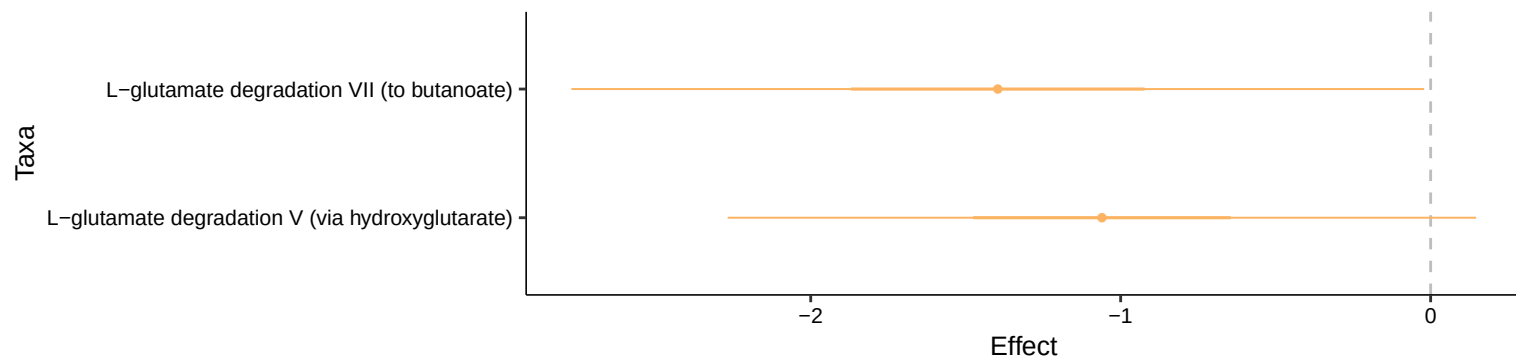
Carboxylic Acid Degradation



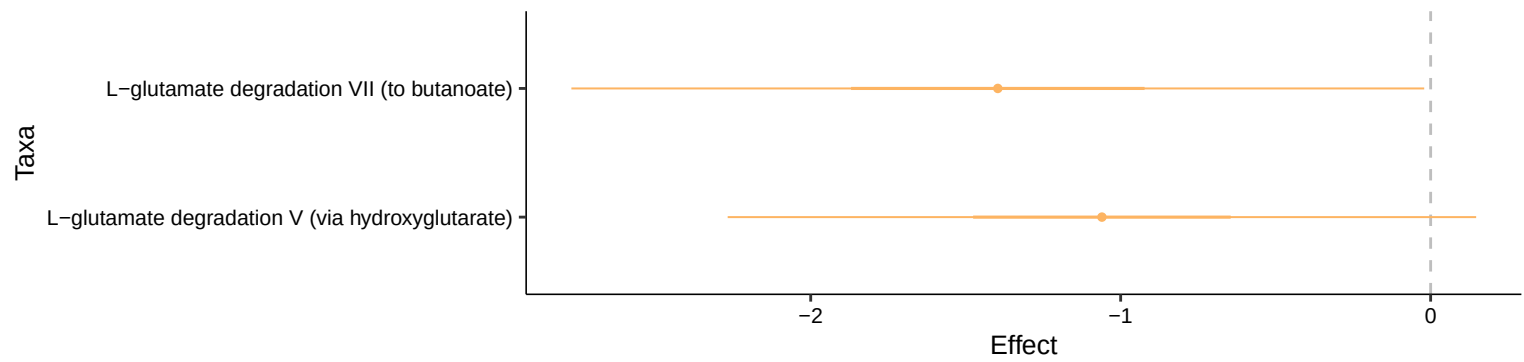
Carbohydrate Degradation&&Alcohol Degradation



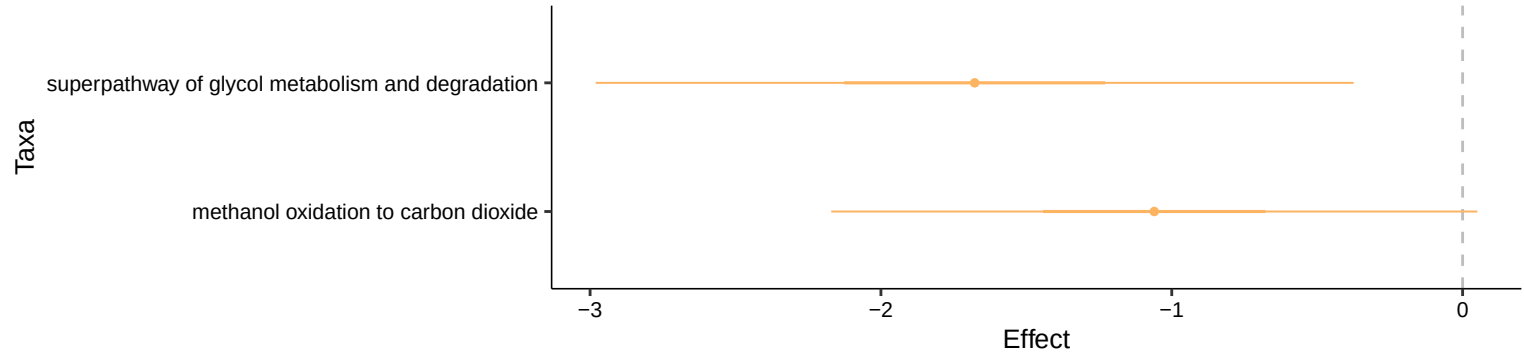
Amino Acid Degradation&&Fermentation



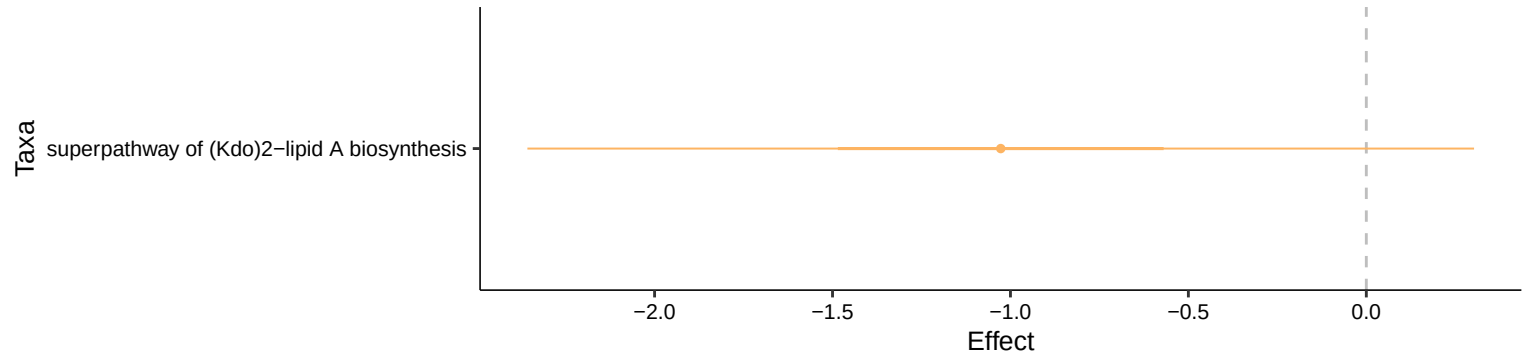
Amino Acid Degradation&&Fermentation



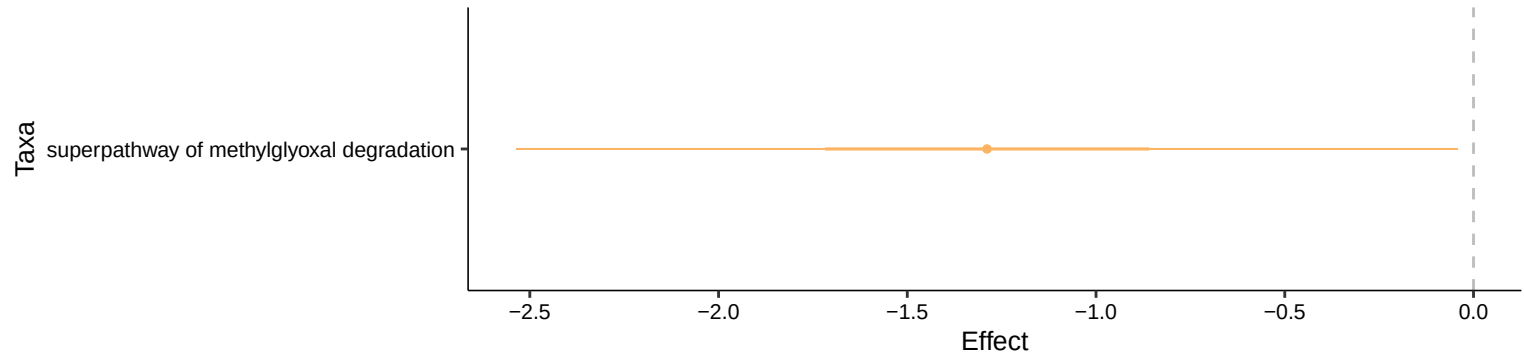
Alcohol Degradation



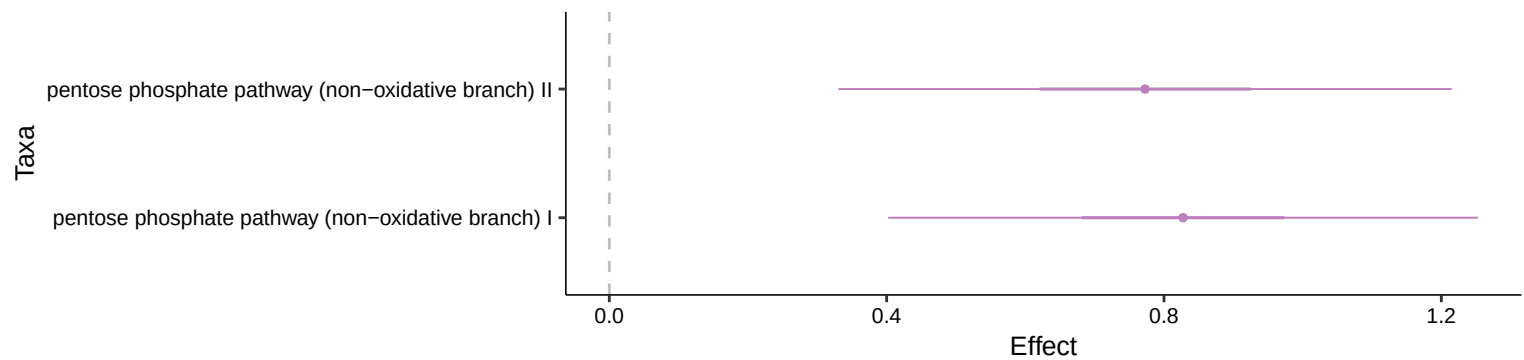
Fatty Acid and Lipid Biosynthesis&&Cell Structure Biosynthesis



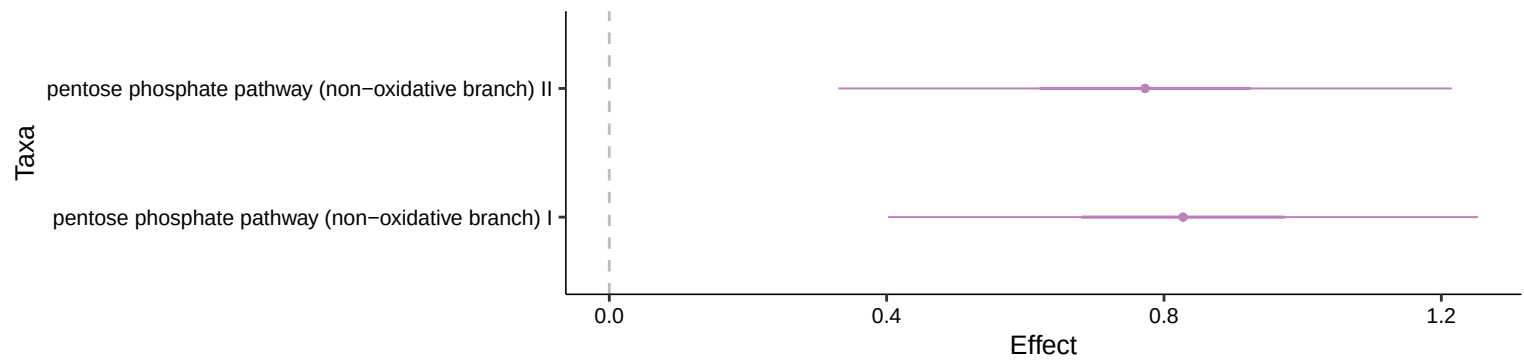
Aldehyde Degradation



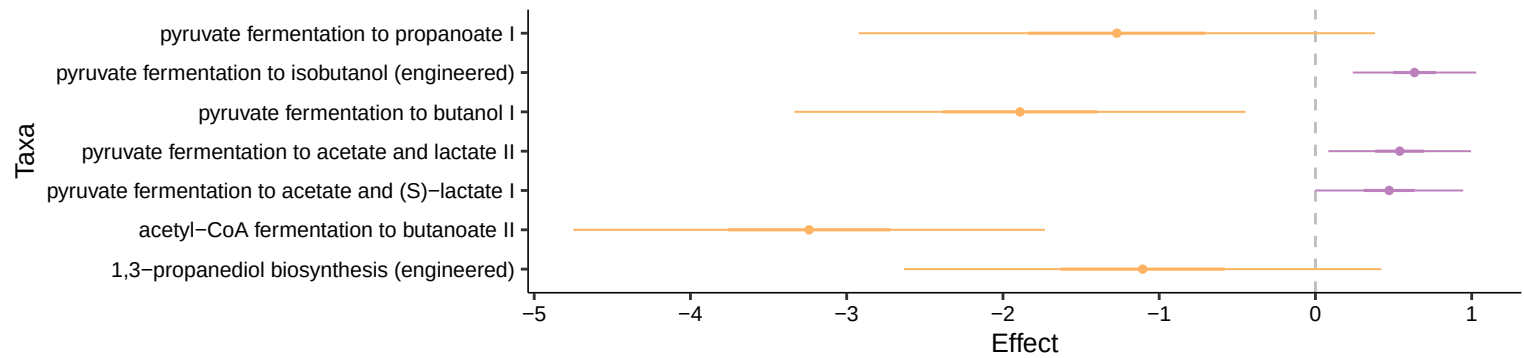
Pentose Phosphate Pathways



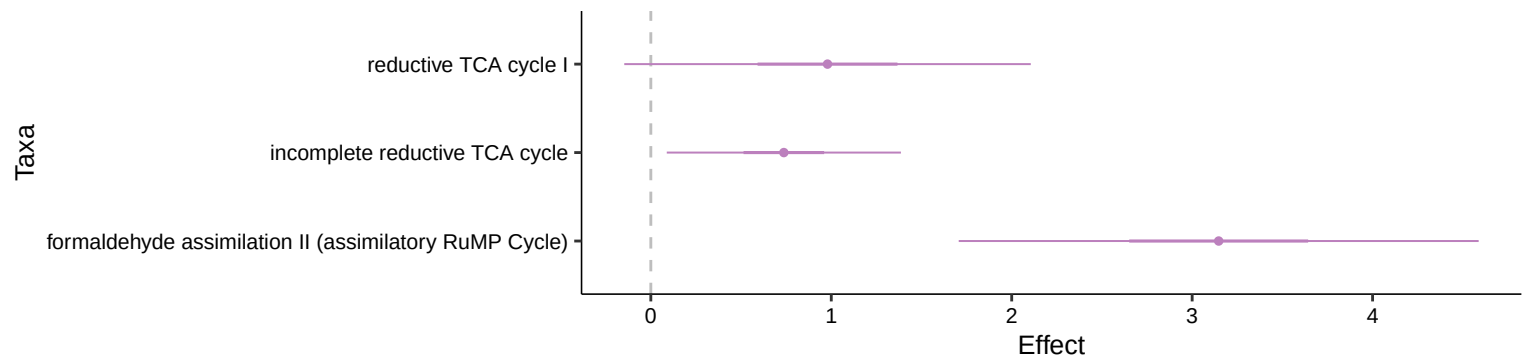
Pentose Phosphate Pathways



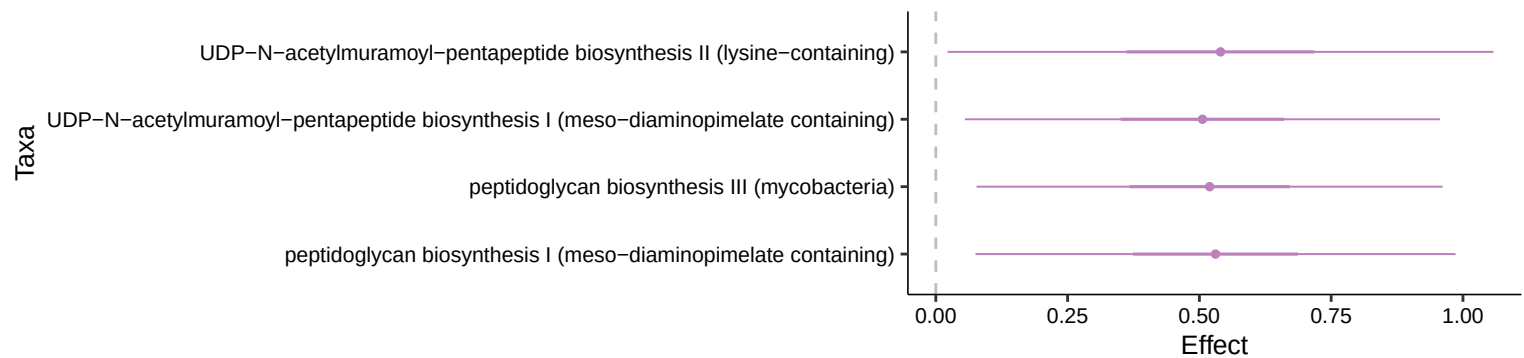
Fermentation



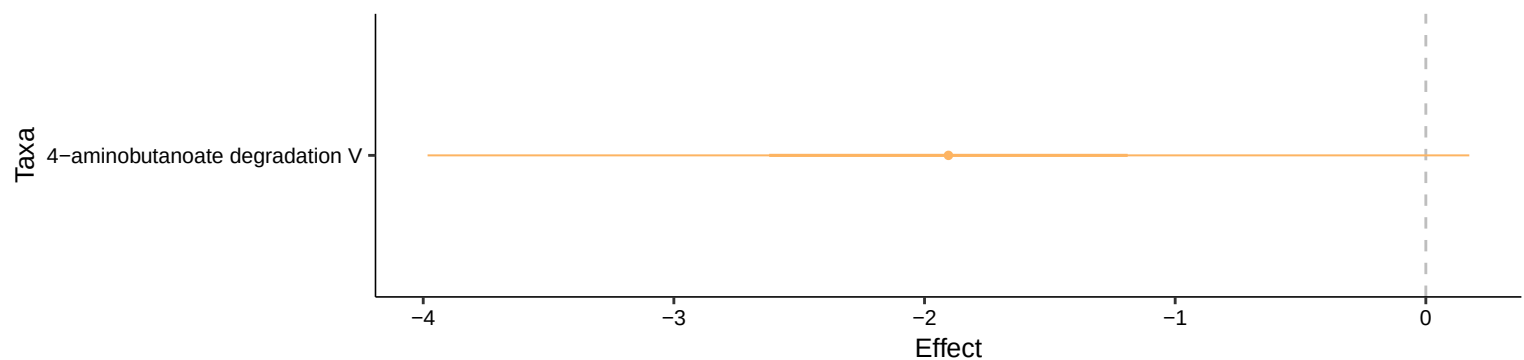
C1 Compound Utilization and Assimilation



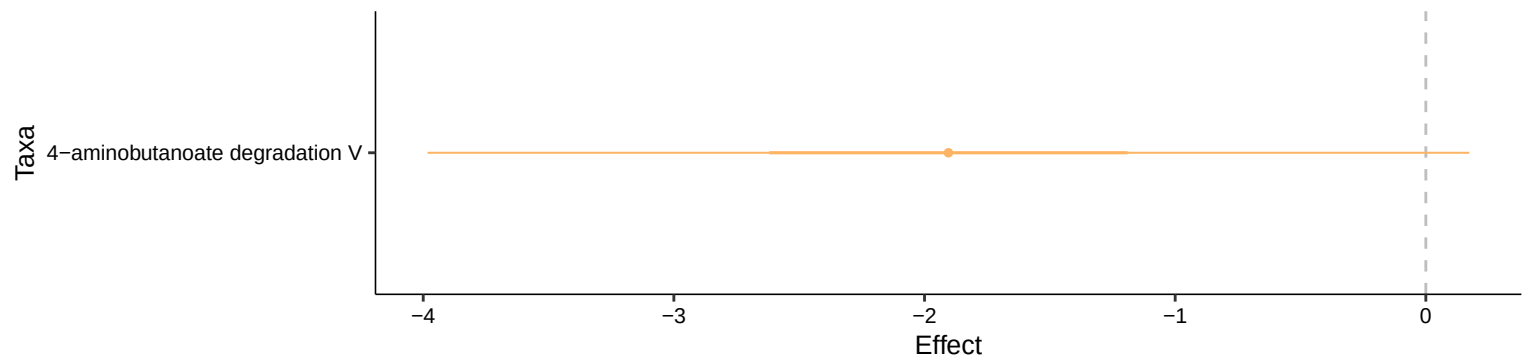
Cell Structure Biosynthesis



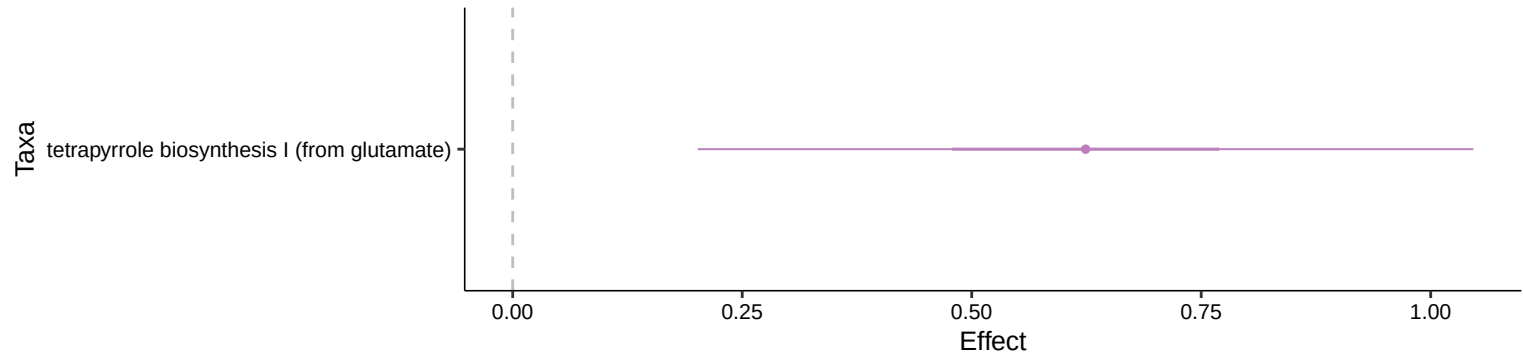
Amide, Amidine, Amine, and Polyamine Degradation&&Fermentation



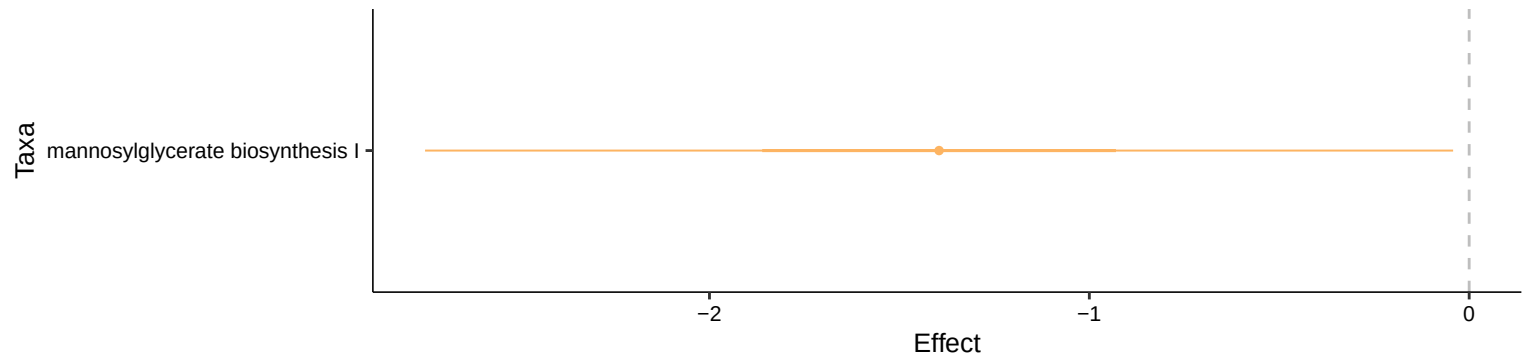
Amide, Amidine, Amine, and Polyamine Degradation&&Fermentation



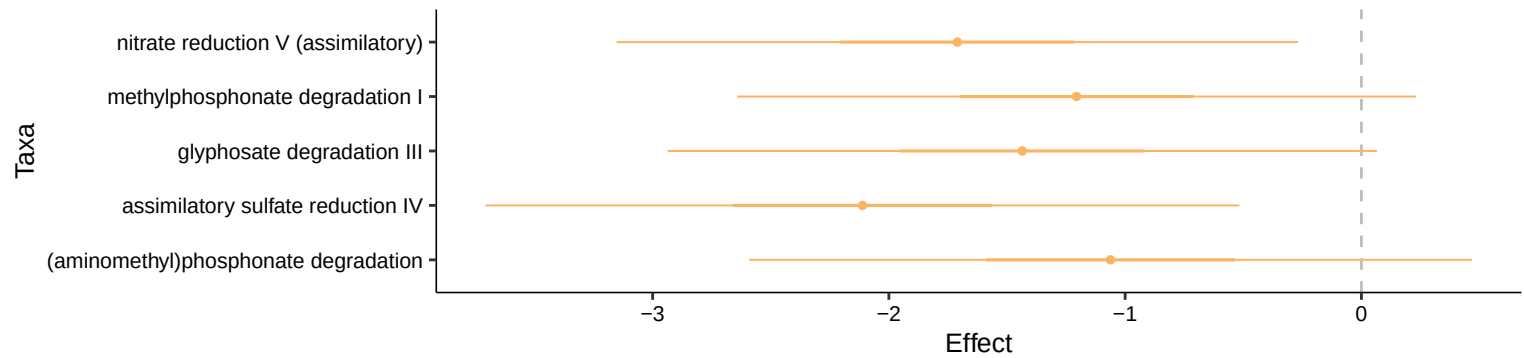
Tetrapyrrole Biosynthesis



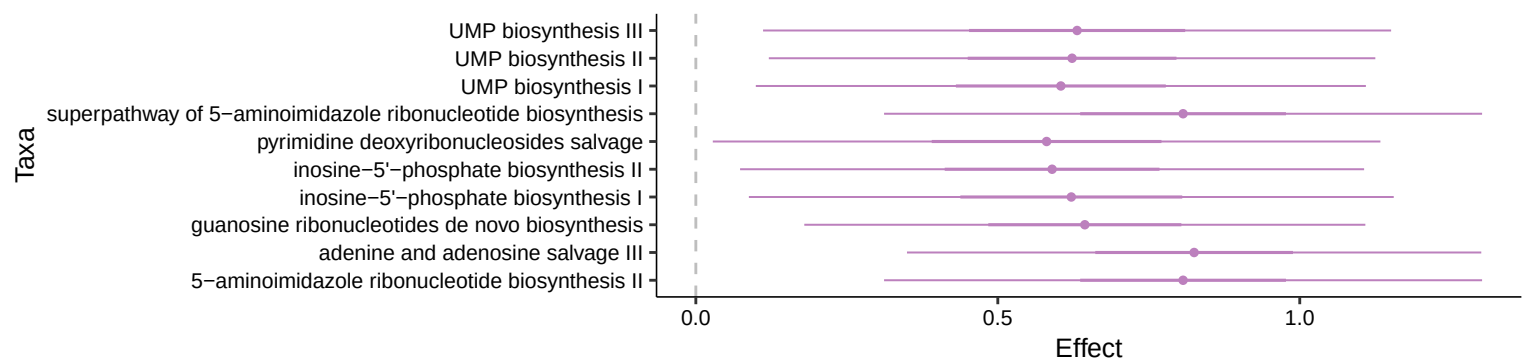
Metabolic Regulator Biosynthesis



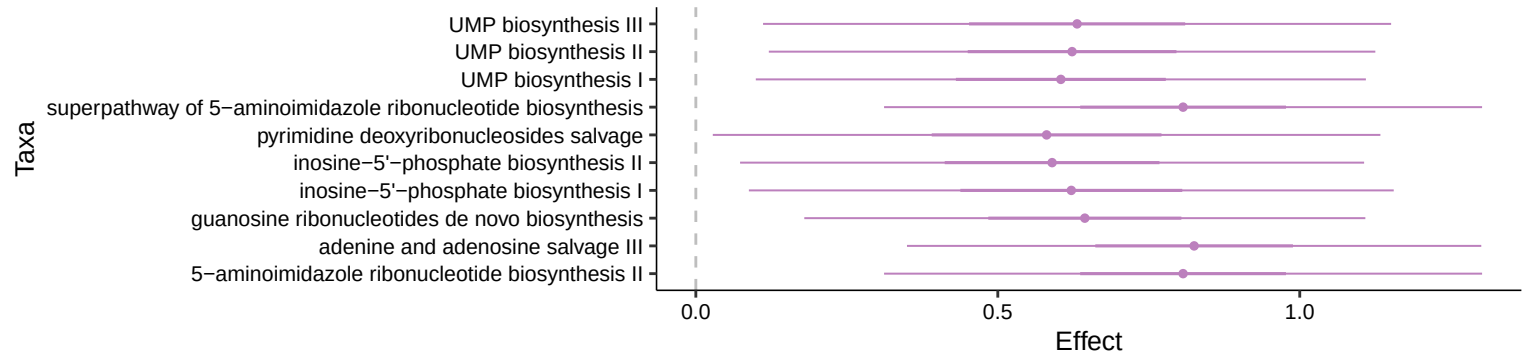
Inorganic Nutrient Metabolism



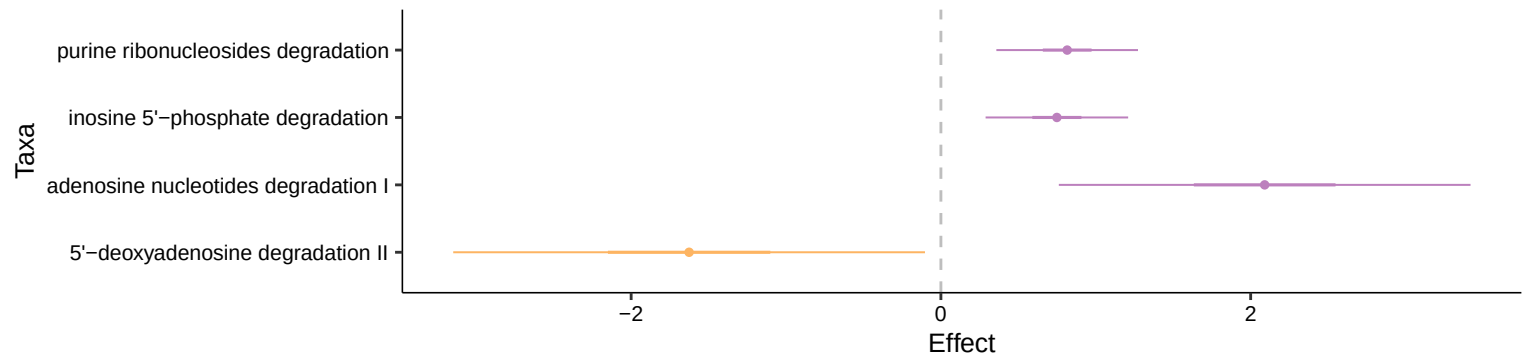
Nucleoside and Nucleotide Biosynthesis



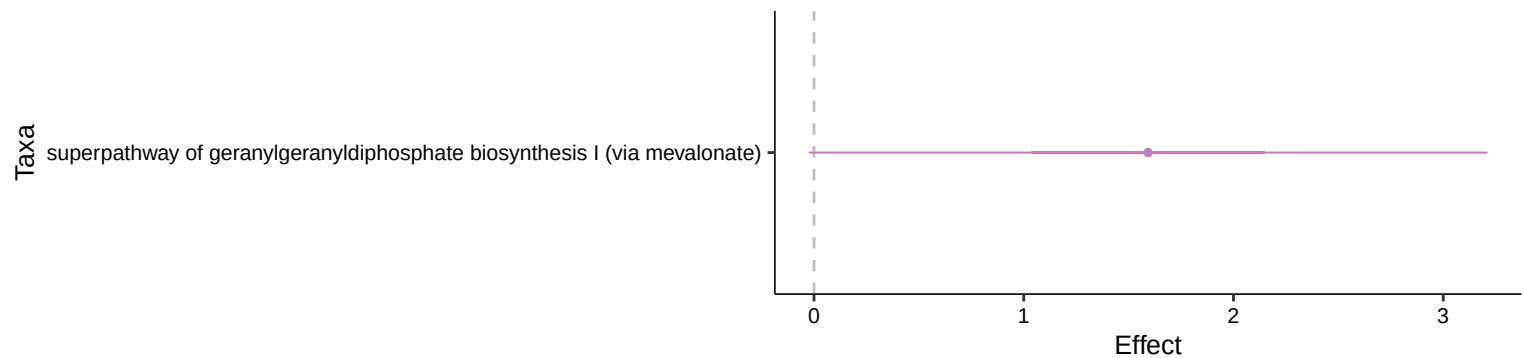
Nucleoside and Nucleotide Biosynthesis



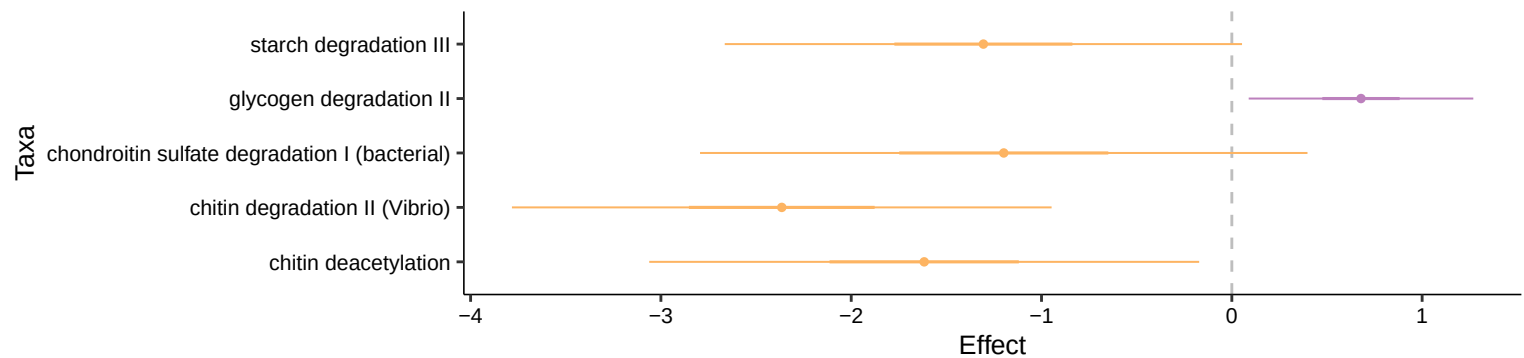
Nucleoside and Nucleotide Degradation



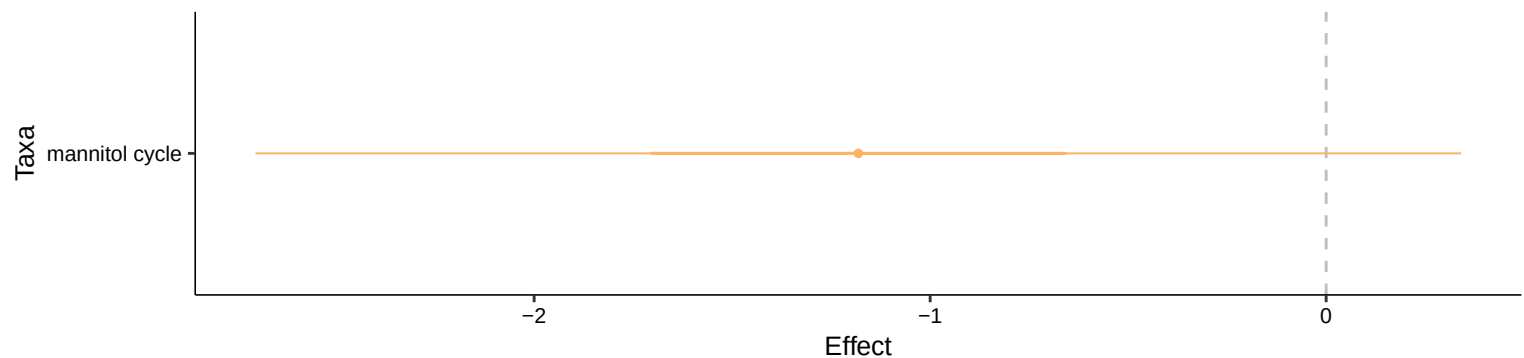
Polyprenyl Biosynthesis&&Secondary Metabolite Biosynthesis



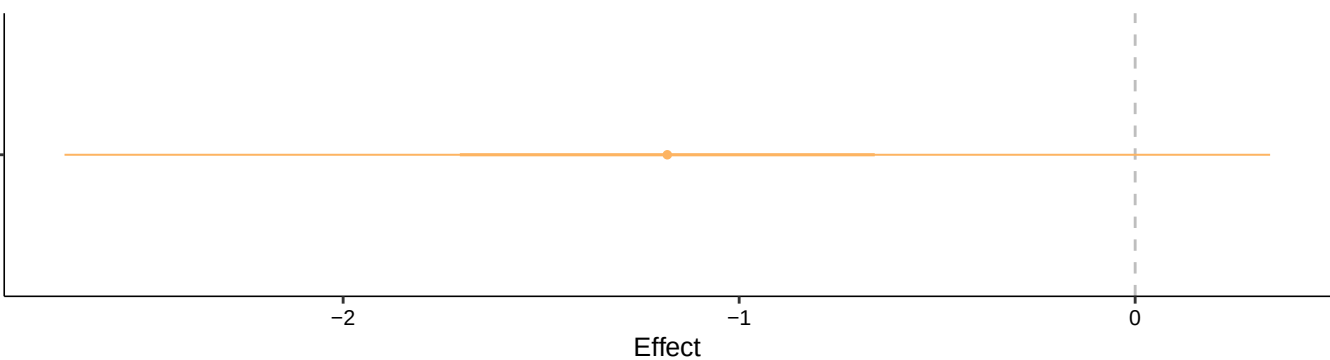
Carbohydrate Degradation&&Polymeric Compound Degradation



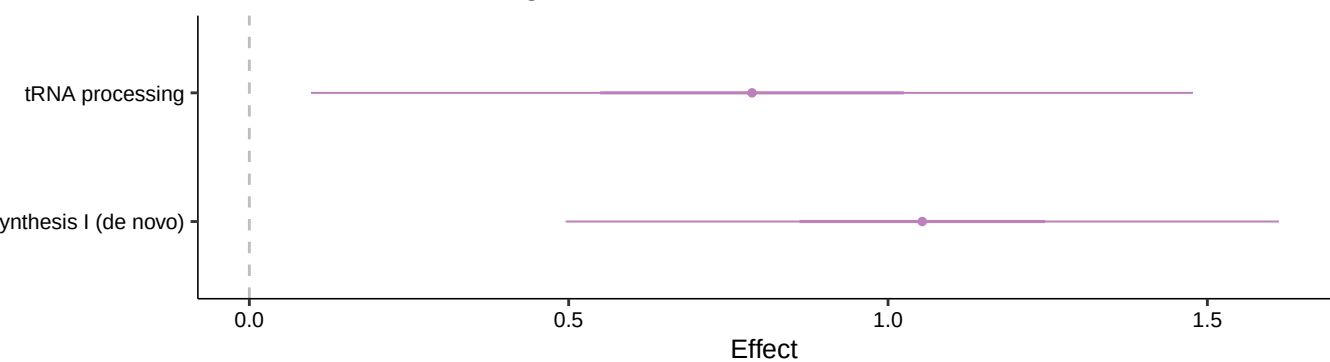
Alcohol Degradation&&Carbohydrate Degradation



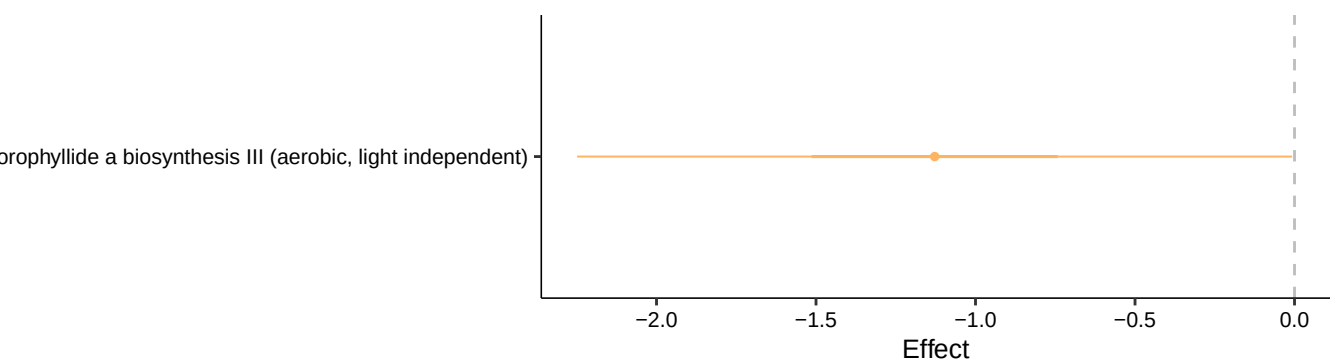
Alcohol Degradation&&Carbohydrate Degradation



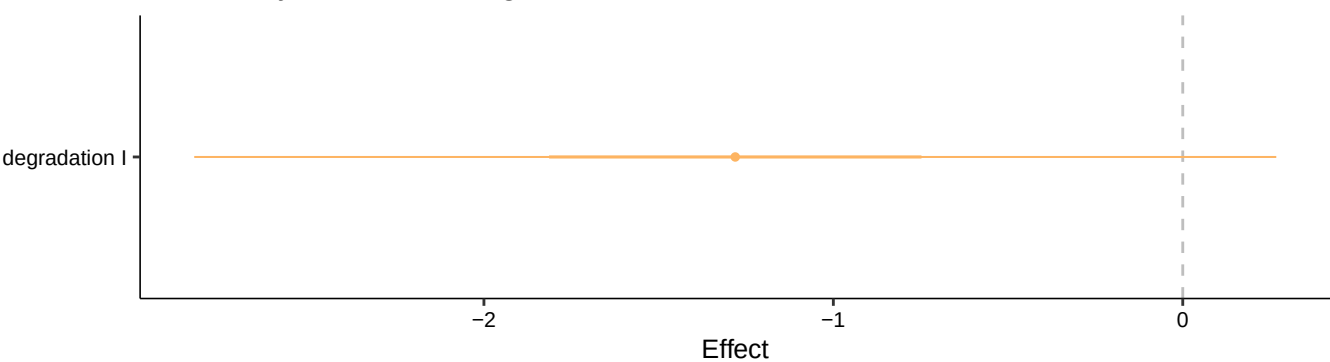
Nucleic Acid Processing



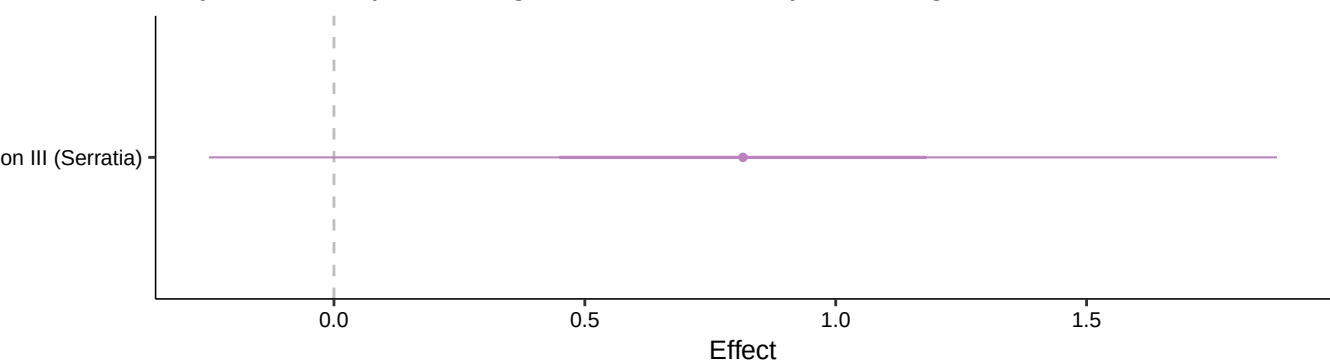
Tetrapyrrole Biosynthesis&&Secondary Metabolite Biosynthesis



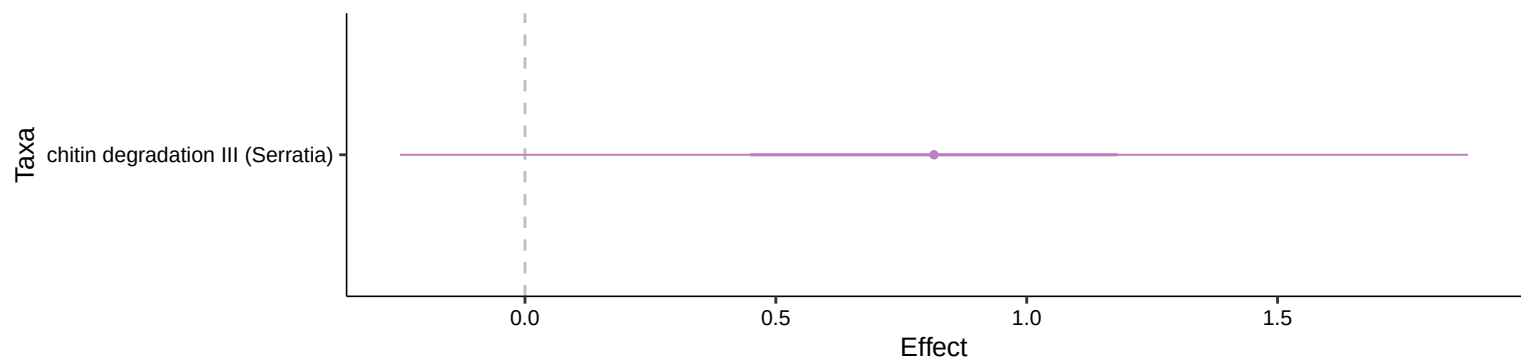
Secondary Metabolite Degradation



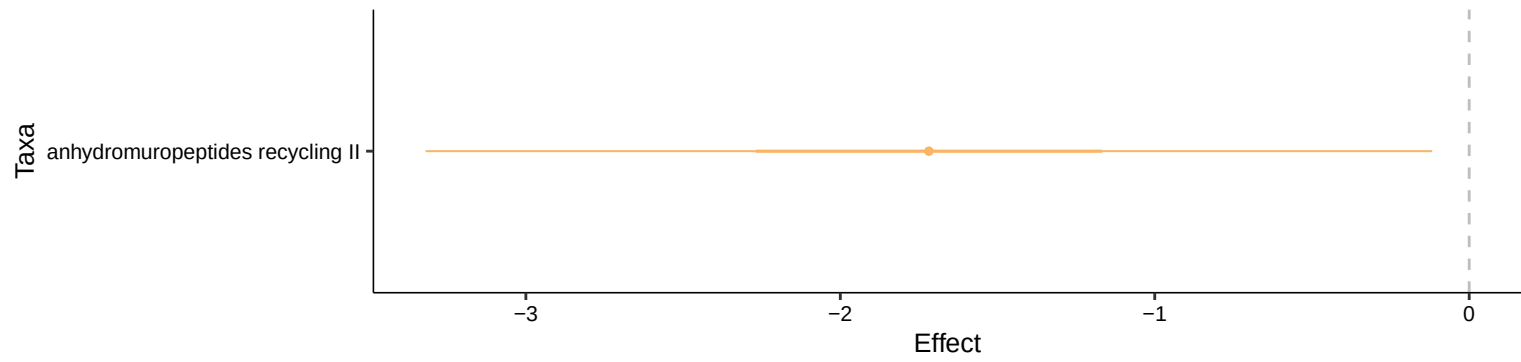
Polymeric Compound Degradation&&Carbohydrate Degradation



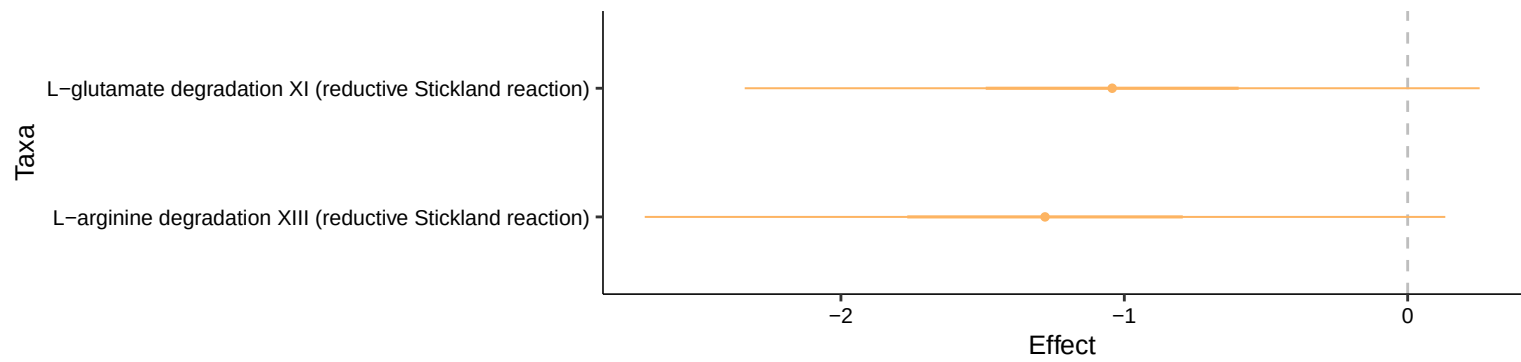
Polymeric Compound Degradation&&Carbohydrate Degradation



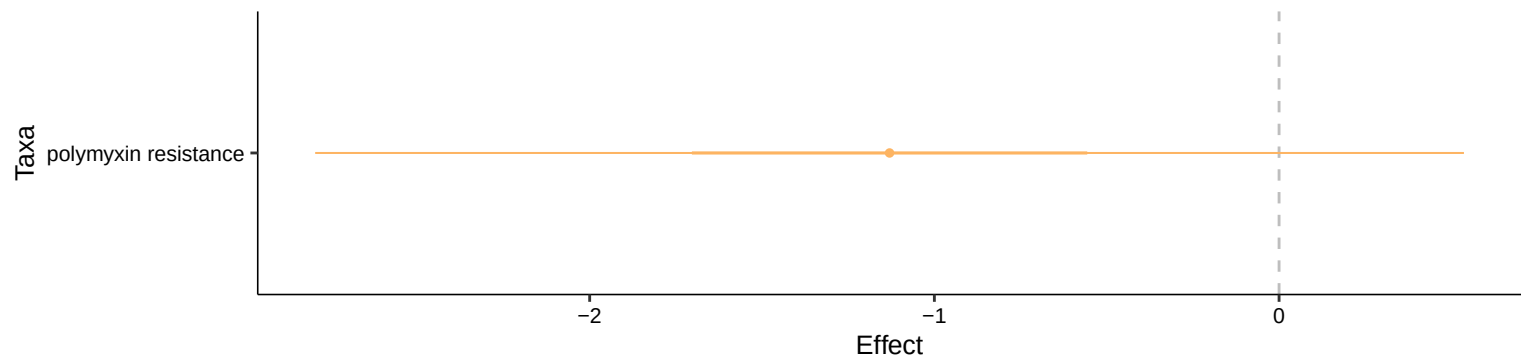
Degradation/Utilization/Assimilation – Other



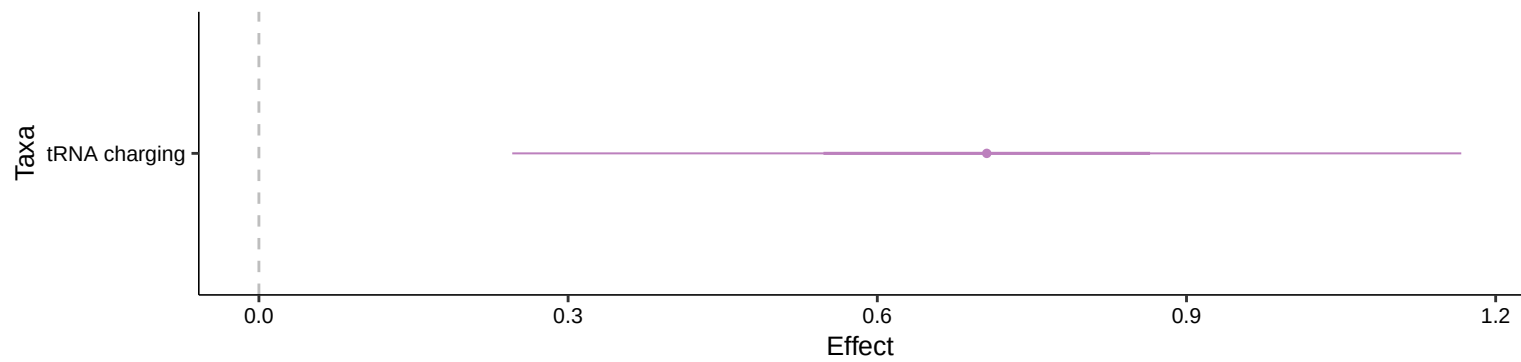
Fermentation&&Amino Acid Degradation



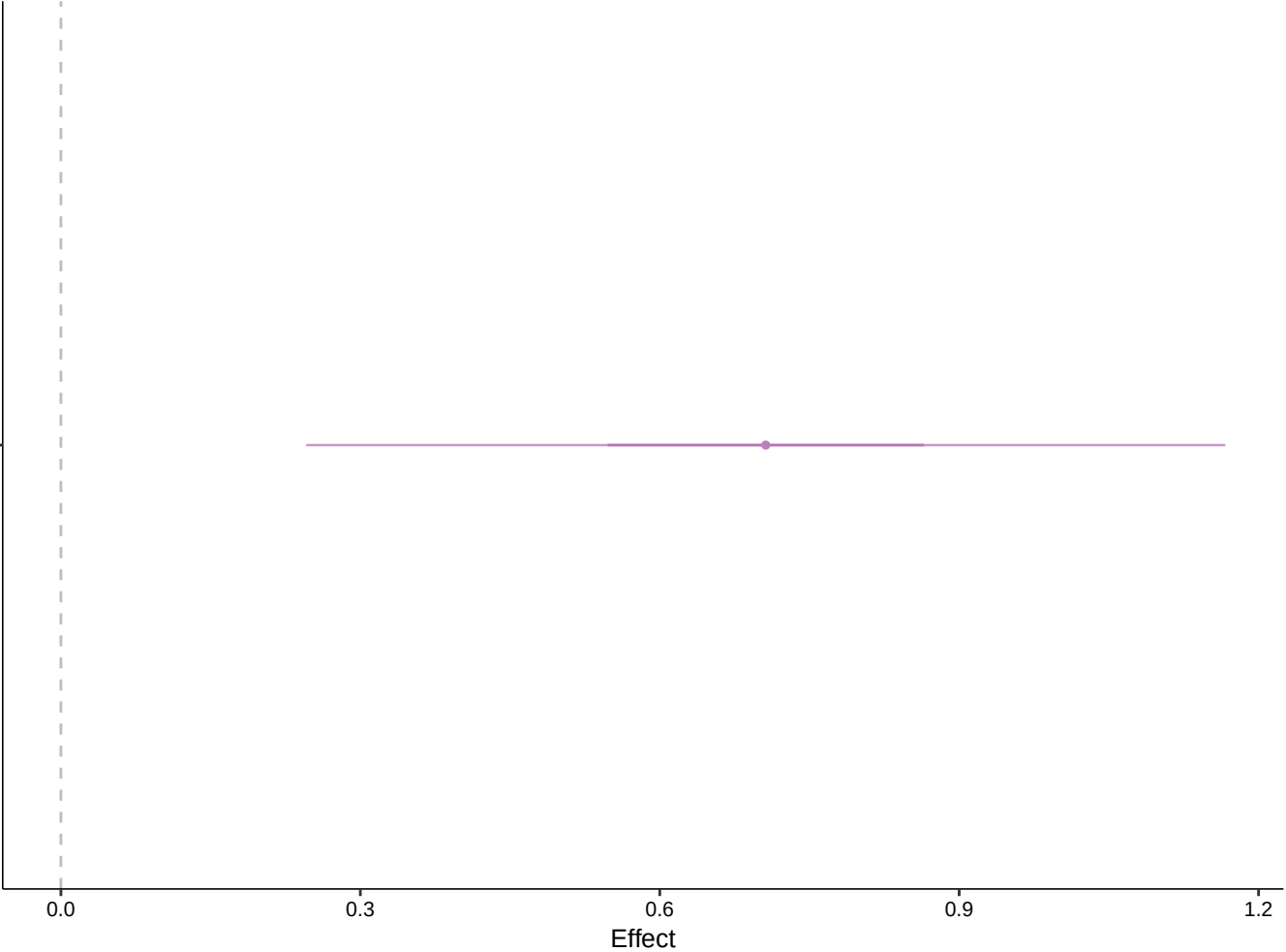
Cell Structure Biosynthesis&&Antibiotic Resistance



Aminoacyl–tRNA Charging



Aminoacyl-tRNA Charging



Amide, Amidine, Amine, and Polyamine Biosynthesis

